MIL-C-306C(ORD)
4 August 1978
SUPERSEDING
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MILITARY SPECIFICATION
CARBON BLACK, POWDERED, DRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers six grades of carbon black for use in ammunition (see 6.1).

1.2 Classification. The carbon black shall be of the following grades, as specified (see 6.2 and 6.4).

Grade A - For use in manufacture of rocket propellant.
Grade B - For use in stabilization of Cyclotol explosives.
Grade C - For use in small arms ammunition propellants.
Grade D - For use in ABL 917, 1362, 1481 and 2434 casting powders.
Grade E - For use in liner material for torpedo warheads, MK 103 and MK 107.
Grade F - For use in propellants for 2.2 KS, Tartar, Standard Arm, Polaris, and PBX (AF) - 108 liner material.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping ,Paper
PPP-B-26 - Bags, Plastic (General Purpose)
PPP-D-705- Drum, Shipping and Storage: Steel, 16 and 30 gallon capacity
PPP-D-723- Drums, Fiber

FSC: 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-L-3061 - Lecithin (For use in explosives)
MIL-C-18164 - Composition D-2
MIL-E-82668 - Explosives, Cyclotol, 25/75 and 29/71 RDX/TNT

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-650 - Explosive, Sampling, Inspection and Testing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multi-level Continuous Sampling Procedures and Tables for Inspections by Attributes

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification
(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D305 - Solvent Extractable Material in Black Pigments, Standard Method of Test for
ASTM-D1483 - Oil Absorption of Pigments by Gardner-Coleman Method, Test for
ASTM-D1506 - Ash Content of Carbon Black, Standard Method of Test for
ASTM-D1509 - Heating Loss of Carbon Black, Standard Method of Test for

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ASTM-D1510 - Iodine Absorption Number of Carbon Black, Test for
ASTM-D1512 - PH Value of Carbon Black, Standard Method of Test for
ASTM-D1513 - Pour Density of Pelleted Carbon Black, Test for
ASTM-D1619 - Sulfur Content of Carbon Black, Standard Methods of Test for
ASTM-D1799 - Sampling Packaged Shipments of Carbon Black, Standard Method for
ASTM-2414 - Dibutyl Phthalate Absorption Number of Carbon Black
ASTM-D3037 - Surface Area of Carbon Black, Standard Methods of Test for
ASTM-E11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 Material. The carbon black shall be a product of high quality, suitable for the purpose intended, and so manufactured as to meet the requirements specified herein.

3.2 Properties. The properties of carbon black shall conform to the requirements in Table I when determined in accordance with the tests of Table II. For Grade A carbon black, the contract or order (see 6.2) shall specify whether the diphenylguanidine requirement shall replace the oil absorption requirement.

3.2.1 Settling (applicable to Grades B and C). The difference in percentage of RDX content of the slurry prepared with the carbon black and Type I, Class 2 Cyclotol, as specified in MIL-E-82668 shall not be greater than two percent from top to bottom of the test sample.

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CHEMICAL AND PHYSICAL PROPERTIES
TABLE I

Properties	Grade A Min	Grade A Max	Grade B Min	Grade B Max	Grade C Min	Grade C Max	Grade D Min	Grade D Max	Grade E Min	Grade E Max	Grade F Min	Grade F Max
Fixed Carbon (%)	97.8*	-	93	-	93	-	NA		94	-	NA	
Moisture (%)	-	2.0*	NA		NA		-	6.0	NA		NA	
Sulfur (%)	-	0.5	NA		NA		-	0.5	NA		NA	
Acetone extract (%)	-	0.5	-	0.5	-	0.5	-	0.5	NA		NA	
Ash (%)	-	0.2	NA		NA		-	0.1	-	2.0	-	0.75
Water soluble Material (%)	-	0.02	-	0.02	-	0.4	-	0.02	NA		NA	
Coarse particles (retained on a US Standard No. 325 Sieve, ASTM-E11 (%))	-	0.2	-	0.2	-	0.2	-	0.1	NA		NA	
Grit (%)	-	0.1	-	0.1	-	0.1	None		NA		NA	
Volatile material (%)	NA		-	7.0	-	7.0	NA		-	6.0	-	6.0
Organic dyes	None		None		None		None		NA		NA	
Oil absorption $\frac{\text{g oil absp.}}{100\text{ g sample}}$	35	45	70	120	60	100	NA		NA		NA	
Diphenylguanidine absorption (%)	-	15	NA		NA		98	-	-	5.4	NA	
Bulk density (g/cm^3)	0.14	0.22	0.14	0.22	0.45	0.55	NA		NA		NA	
Compatibility (cm^3/g)	NA		-	2.0	-	2.0	NA		NA		NA	
Iodine absorption $\frac{mgI_2}{g\text{ sample}}$	NA		25	35	25	35	NA		NA		NA	
Dibutyl Phthalate absorption number	NA		NA		NA		NA		60	70	NA	
Acidity (pH)	NA		NA		NA		2.0	5.0	4.0	7.0	7.0	9.5
Yield value $\frac{\text{dyne}}{cm^2}$	NA		8	25	8	25	NA		NA		NA	
Settling	NA		see 3.2.1		see 3.2.1		NA		NA		NA	
Surface area $\frac{m^2}{g}$	NA		NA		NA		NA		87.0	126.0	NA	

*For Army purchases only, the fixed carbon shall be 96.8 percent min. and the moisture 3.0 percent max.

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3.3 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.2 and 4.3).

3.4 Workmanship. The carbon black shall be processed in a manner that will yield the high quality product necessary to meet the requirements of this specification. The carbon black shall be free of foreign matter such as wood, paper, metal and dirt when determined as specified in 4.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance (regular production) inspection (see 4.4).

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of twenty (20) pounds of carbon black obtained by sampling as described in 4.4.3. The sample shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The first article sample shall be subjected to the inspections specified in Table II (see 4.4.3) for the applicable grade of carbon black.

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4.3.3 Rejection. If the sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. For the material covered by this specification, a lot shall consist of one or more batches of carbon black produced by one manufacturer, at one plant, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions and to be offered for acceptance at one time. Each batch shall consist of that quantity of carbon black that has been subjected to the same unit chemical or physical process. The lot shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-306C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Drums, Metal or fiber, prior to closing				NA
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Bag missing, torn or punctured		0.40%	5.1	Visual
102.	Bag not securely closed		0.40%	5.1	Visual/Manual
103.	Drum not as specified		0.40%	5.1	Visual
Minor:	None defined.				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-306C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Drums, metal or fiber, after closing				NA
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Gross weight of drum in excess of that specified.		0.40%	5.1	Scale
102.	Drum improperly closed.		0.40%	5.1	Visual
103.	Drum cut, scraped, dented or damaged so as to affect serviceability.		0.40%	5.1	Visual
104.	Marking illegible or incorrect.		0.40%	5.2	Visual
Minor:	None defined				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-306C

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Bags, paper or plastic, after filling				NA
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Weight incorrect.		0.40%	5.1	Scale
102.	Bag not as specified.		0.40%	5.1	Visual
103.	Bag improperly closed.		0.40%	5.1	Visual
104.	Bag cut, torn, punctured.		0.40%	5.1	Visual
105.	Bag improperly marked.		0.40%	5.1	Visual
Minor:	None defined				

NOTES:

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4.4.3 Sampling and testing. Three (3) containers shall be selected at random from each lot. Approximately a one (1) kilogram sample of carbon black shall be withdrawn from each container using ASTM Procedure D1799. Subsamples for each sample shall be subjected to the applicable inspections specified in Table II. If any subsample fails to meet the applicable requirements (paragraph 3), the lot shall be rejected. The classification for the inspections shall be as given in Table II.

TABLE II

Inspection/Requirement (see 3.2)	Defect Classification	Test Paragraph
Fixed carbon	Major	4.5.1
Moisture	Major	4.5.2
Sulfur	Major	4.5.3
Acetone extract	Major	4.5.4
Ash	Major	4.5.5
Water soluble material	Major	4.5.6
Coarse particles	Major	4.5.7
Grit	Major	4.5.8
Volatile material	Major	4.5.9
Organic dyes	Major	4.5.10
Oil absorption	Major	4.5.11
Diphenylguanidine absorption	Major	4.5.12
Bulk density	Major	4.5.13
Compatibility	Major	4.5.14
Iodine absorption	Major	4.5.15
Dibutyl Phthalate absorption number	Major	4.5.16
Acidity	Major	4.5.17
Surface area	Major	4.5.18
Yield value (applicable for First Article Inspection only)	Major	4.5.19
Settling test (applicable for First Article Inspection only)	Major	4.5.20
Workmanship	Major	4.5.21

4.4.4 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.3).

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4.5 Test methods and procedures (see 6.5). The inspections in 4.5.1 through 4.5.21 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

PRECAUTION - This specification covers sampling and testing of chemically toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Fixed carbon

4.5.1.1 Fixed carbon applicable to Grade A only. The percentage of fixed carbon shall be determined by subtracting the percentage of moisture (4.5.2) and ash content (4.5.5) from one hundred.

4.5.1.2 Fixed carbon applicable to Grades B, C and E only. The percentage of fixed carbon shall be determined by subtracting the percentage of volatile matter (4.5.9) from one hundred.

4.5.2 Moisture. The moisture content of the sample shall be determined in accordance with ASTM-D1509.

4.5.3 Sulfur. The sulfur content of the sample shall be determined in accordance with ASTM-D1619.

4.5.4 Acetone extract. The quantity of acetone extractable material shall be determined in accordance with ASTM-D305. Retain the acetone extract for the examination for organic dyes as specified in 4.5.10.

4.5.5 Ash. The ash content of the sample shall be determined in accordance with ASTM-D1506.

4.5.6 Water soluble material. Transfer an accurately weighed portion of approximately 5g of the sample to a 400mL beaker. Add 150mL of distilled water, mix well and boil for 15 minutes. Cool the mixture and quantitatively transfer it to a 200mL volumetric flask. Dilute to the mark with distilled water, mix thoroughly and allow to settle. Decant approximately one half of the supernatant liquid and filter through a dry filter paper. Discard the first 10mL of the filtrate and transfer 100mL of the clear filtrate to a tared dish. Evaporate to dryness on a steam bath (or hot plate), dry for 1 hour in an oven at 100° ± 5°C, cool in a desiccator and weigh. Calculate the weight of residue as percentage of water soluble material in the sample on a moisture-free basis. Retain the remaining filtrate for examination for organic dyes as specified in 4.5.10.

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4.5.7 Coarse particles. Prepare a stock solution of dispersing agent in water using as high a concentration of dispersing agent as possible without losing fluidity, and filter through coarse filter paper. (Triton 720, Darvan No. 1, Nacconol, or any other dispersing agent suitable for carbon black may be used.) Place a 10g portion of the sample in a 600mL beaker. Add enough dispersing agent solution to form a heavy paste, and mix well to thoroughly incorporate the carbon black. Dilute the mixture with water to a volume of approximately 300mL, and then pour through a tared U. S. Standard No. 325 sieve (45um). Wash the material on the sieve with a steady, gentle stream of water which first has been passed through a U. S. Standard No. 325 sieve. Using a small camel's hair brush, brush the material through the sieve, running water through at the same time. Continue this operation until no more carbon black passes through the sieve. Once or twice during the screening, it may be helpful to remove the sieve from under the running water, add a drop or two of the stock dispersing agent solution, thoroughly spread it over the sieve with the brush, and continue the washing and brushing. Dry the sieve in an oven at 100° to 105°C, for 1 hour, cool and weigh the residue. Calculate the weight of residue as percentage of coarse particles in the sample on a moisture-free basis. Retain the residue for the determination of grit (see para. 4.5.8).

4.5.8 Grit. Transfer the residue (if any) obtained as specified in paragraph 4.5.7 to a piece of smooth white paper. Gently rub the residue with the finger until the paper is no longer blackened by the remaining dispersible carbon black. Transfer the grit particles to a tared weighing dish, weigh and calculate the weight of the particles as percentage of grit in the sample on a moisture-free basis.

4.5.9 Volatile content

4.5.9.1 Apparatus

1) Electric furnace - capable of temperature regulation of + 20°C at 950°C and equipped with a thermocouple activated indicating pyrometer.
2) Platimum crucible - 20mL capacity, equipped with insert form cover with 1.0mm hole in top.
3) Interval timer
4) Oven - capable of temperature regulation at + 1°C at 105°C.
5) Analytical balance - having a sensitivity of 0.1mg.
6) Desiccator - with indicating desiccant.

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4.5.9.2 Procedure for obtaining tare weight. Fill the platinum crucible with a non-test sample of carbon black to a level of not more than 2mm below the cover line. Insert the crucible cover in place and heat the covered crucible and contents in the electric furnace for 10 minutes at 950 ± 20°C. Remove from the furance, place in a desiccator and allow to cool. Remove the covered crucible and contents from the desiccator and empty the carbon black from crucible. Brush out any carbon black adhering to the walls and immediately weigh the crucible and cover to the nearest 0.1mg.

If the crucible and cover had recently been used for determining volatile loss of carbon black and were brushed clean and placed in a desiccator immediately after use, the tare weight may be obtained without refiring with carbon black.

4.5.9.3 Procedure for sample determination. Dry an adequate sample for 1 hour min. at 105 ± 5°C as described in ASTM-D1509. (This step is not applicable for Grade E carbon black). Fill the tared platinum crucible with the carbon black, and firmly tamp the material. The level of sample shall not be more than 2mm below the cover line if the sample is expected to have volatile content below 6%, and not more than 10mm below the cover line if expected volatile content is 6% or above. Fit the crucible cover firmly in place and weigh the crucible, cover and contents to the nearest 0.1mg. Ignite the covered crucible and contents in the electric furnace for exactly 7 minutes at a temperature of 950 ± 20°C. Immediately remove the covered crucible and contents from the furnace, place in a desiccator and allow to cool to room temperature. Weigh the covered crucible and contents to the nearest 0.1mg and calculate the percent volatile content as follows:

$$V = \frac{(B-C)}{(B-A)} \times 100$$

where:

V = Volatile content in percent
A = Weight of crucible and cover in grams
B = Weight of crucible, cover and sample before ignition, in grams
C = Weight of crucible, cover and sample after ignition, in grams.

4.5.10 Organic dyes. Visually examine the acetone and water extracts (see 4.5.4 and 4.5.6), using normal laboratory illumination, for the presence or absence of organic dyes.

4.5.11 Oil absorption. The oil absorption of the sample shall be determined in accordance with ASTM-D1483.

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4.5.12 Diphenylguanidine absorption

4.5.12.1 Reagents

4.5.12.1.1 Diphenylguanidine solution. Prepare the 0.01N diphenylguanidine solution as follows: Dissolve 2.00g of chemically pure diphenylguanidine in 1 liter of ethanol. The ethanol shall be of such quality that when it is diluted with an equal volume of water, no more than a slight turbity shall be produced. A milkiness will ruin the solution for titrating. The solution will be slightly less than 0.01N.

4.5.12.1.2 Bromophenol blue - methyl red indicator. Prepare the bromophenol blue - methyl red indicator as follows: Place 0.066g of bromophenol blue and 0.166g of methyl red in a beaker and add 100mL of ethanol. Heat the mixture almost to the boiling point to dissolve the crystals, cool and filter.

4.5.12.2 Procedure. Transfer an accurately weighed portion of 2g of the sample to a 250mL Erlenmeyer flask and add a measured portion of 100mL of approximately 0.01N diphenylguanidine solution in alcohol. Place a measured portion of 100mL of 0.01N diphenylguanidine solution in alcohol in another 250mL Erlenmeyer flask for a blank determination. Stopper both flasks tightly, place on a shaking machine, and shake for 2 hours. Remove the flasks from the machine. Remove all, or nearly all, of the suspended carbon by filtration. If this is impractical, because of clogging of the filters, centrifuge the mixture.

If the filtration is used, filter the solution containing the carbon black through filter paper, shaking the flask thoroughly before transferring the material to the filter paper so that the carbon black will be thoroughly mixed with the solution. The carbon black forms a pad which tends to prevent the very fine particles from passing through the filter paper. Return the filtrate to the filter until the filtrate is relatively clear. Place a suitable cover over the funnel during the filtration and keep the funnel in a place where there are no air currents, so as to minimize evaporation of the alcohol. Treat the blank solution in the same way, returning the filtrate to the funnel the same number of times as in the sample solution so that the evaporation of alcohol will be approximately the same in each case. By means of a pipette, transfer 50mL of each filtrate to separate 250mL Erlenmeyer flasks. Add 9 drops of bromophenol blue-methyl red indicator to each flask, and titrate rapidly with 0.01N hydrochloric acid from a burette until the indicator begins to change color. Then slow down the addition to drop by drop rate. Due to the dilution of reagents, the indicator will not show a sharp change of color.

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The color in the diphenylguanidine solution will be green, and the first change in color will be toward a red. The changes will be through various shades until a deep red is reached and the green has entirely disappeared.

If centrifuging is used, centrifuge out most of the carbon by pouring both blank and sample suspension into 100mL centrifuge tubes, stoppering and centrifuging at high speed for 30 minutes or more until the supernatant is essentially clear (For Grade D carbon black, it may be necessary to centrifuge for three (3) hours). Pipette 50 mL of the supernatant liquid from each centrifuge tube into separate beakers, and titrate each with 0.01N acid as described above. If the remaining suspended carbon prevents titration with the indicator for detection of the end point use a quantitative titrimeter fitted with glass and calomel electrodes. The pH of the mixed indicator endpoint is 4.

Calculate the percentage of diphenylguanidine present in the solution absorbed by 1g of the carbon black as follows:

$$\text{Percentage of diphenylguanidine absorbed} = \frac{(A-B)\ 100}{A}$$

where:

A = mL of 0.01N hydrochloric acid used for blank
B = mL of 0.01N hydrochloric acid used for sample

4.5.13 Bulk density. The bulk density shall be determined in accordance with ASTM-D1513, except that the pour density shall be converted to g/cm3.

4.5.14 Compatibility. The compatibility shall be determined in accordance with MIL-STD-650, Method 504.1 at a temperature of 100° ± 0.5°C, except that the test samples shall be prepared as follows:

a. The first sample shall contain one (1) gram of Cyclotol explosive, Type I, Class 2 as specified in MIL-E-82668.
b. The second sample shall contain one (1) gram of carbon black.
c. The third sample shall contain an intimate mixture of one (1) gram of Cyclotol and one (1) gram of carbon black.

Also, the calculation shall be:

$$\text{Gas due to reactivity, } cm^3/g = \frac{C - (A+B)}{2}$$

where:

C = amount of gas evolved by the 50/50 mixture of Cyclotol and carbon black.
A = amount of gas evolved by the Cyclotol sample.
B = amount of gas evolved by the carbon black sample.

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4.5.15 Iodine absorption. The iodine absorption of the sample shall be determined in accordance with ASTM-D1510.

4.5.16 Dibutyl phthalate absorption number. The dibutyl phthalate absorption number of the sample shall be determined in accordance with ASTM-D2414.

4.5.17 Acidity. The acidity of the sample shall be determined in accordance with ASTM-D1512.

4.5.18 Surface area. The surface area of the sample shall be determined in accordance with ASTM-D3037.

4.5.19 Yield value

4.5.19.1 Preparation of slurry. Prepare a batch of Type I, Class 2 Cyclotol (see 3.2.1), with a carbon black, in a gallon wide mouth, steam heated kettle. The kettle shall be equipped with a Palo impeller agitator (see 6.6) containing an air driven motor. Place aluminum foil around the top of the kettle to prevent drafts from cooling the surface of the explosive.

The temperature of the mix shall be 95 + 1°C. All the components shall be accurately weighed to 0.1g. Firstly, add 1800g of Cyclotol to the kettle and melt. Then add the remaining components to the kettle in the following order: 16.7g of Composition D-2 (in accordance with MIL-C-18164), 1.8g of lecithin (in accordance with MIL-L-3061), except that the requirements for 93 + 10 penetration shall not apply) and 54g of carbon black. After the addition of carbon black, mix the batch for exactly 15 minutes at 800 RPM.

4.5.19.2 Procedure. Viscosity measurements, of the mix, are then made at 95 + 0.5°C with a Brookfield Viscometer, Model RVT or equivalent, using a preheated (95°C) spindle #1. The measurements are made at increasing RPM's of 0.5, 1.0, 2.5, 5.0, 10.0, 20.0, 50.0 and 100.0 respectively. Then, the measurements shall be retraced. The last two measurements at 1.0 and 0.5 RPM, at a decreasing rate of shear, are used to calculate yield value. Use the following equation to calculate the yield value (see 6.7):

$$Y = \frac{2\ r_1\ (V_1^{\ 1} - V_2^{\ 1})}{100}$$

where:

Y = Yield value (dynes/cm^2)
r_1 = 0.5 RPM
(r_2 = 1.0 RPM)
$V_1^{\ 1}$ = apparent viscosity at r_1 (centipoise)

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V_2^1 = apparent viscosity at r_2 (centipoise)

2 = proportionality factor between r_1 and r_2

Retain the Cyclotol mix for the settling test (see 4.5.20).

4.5.20 Settling test. Immediately after the completion of the yield value test, pour the Cyclotol mix into two preheated (95°C) aluminum molds. The molds, equipped with aluminum foil risers, shall be four inches in height and have a one inch inside diameter. Pour the explosive into the riser to a depth of one inch. Place the molds, containing the explosive, into an explosion-proof oven at 95 ± 1°C for five hours. Remove the molds and contents from the oven and allow the samples to solidify at ambient laboratory temperature. Saw off the riser section. Analyze the top one inch (under the riser) and the bottom one inch of each stick, for RDX as follows:

4.5.20.1 Procedure for determining TNT, content. Reduce the sample, using a wooden mortar and pestle, through a No. 20 mesh sieve (U. S. Standard). Accurately weigh an approximately two (2)g sample to the nearest 0.1mg into a tared, sintered glass, medium porosity, filtering crucible fitted with a filter paper disc. Cut the filter paper disc from Fisher 9-795 or equivalent semi-crimped, rapid filter paper. The filter paper shall be of the same diameter and make a tight seal with the bottom of the crucible. Quantitatively transfer the sample to a one hundred and fifty (150) milliliter (mL) beaker (see 6.8). Add twenty (20) mL of reagent grade benzene saturated with RDX. (The benzene should become saturated with RDX at room temperature within a minimum of twelve (12) hours and filtered just prior to use). Heat the beaker on an oscillating hot plate for thirty (30) minutes at a temperature of 70° to 75°C. Disintegrate the sample by oscillating or if necessary by use of a stirring rod. Upon removal from the hot plate, wash down the sides of the beaker with approximately ten (10) mL, RDX sat'd. benzene. Cool the sample to room temperature. Quantitatively transfer the sample to the original tared filtering crucible in a Fisher filtrator, No. 9-788, equipped with crucible holder, Fisher No. 8-285, connected to a water aspirator. Transfer the sample, with the aid of a wash bottle containing benzene saturated with RDX. Filter the sample. Rinse the beaker three (3) additional times with 15mL portions of benzene - RDX solution from the wash bottle. The rinsings shall be added to the crucible. The contact time with the crucible residue shall be five (5) to ten (10) seconds. Do not disturb the filter paper when stirring the crucible and contents during washings. Wash down the sides of the crucible with a fourth rinsing. Aspirate for thirty (30) seconds after the last washing. Dry the crucible and its contents in an oven, at 95° + 5°C, for thirty (30) minutes. Cool in a desiccator for thirty (30)

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minutes, weigh and calculate the percent TNT, including wax and lecithin present, according to the following formula.

$$\text{Percent TNT} = \frac{(A-B)\times 100}{C}$$

where:

A = weight of crucible plus sample, in grams
B = weight of crucible plus first residue, in grams
C = weight of sample, in grams

Retain the residue for determination of carbon black (4.5.20.2)

4.5.20.2 Procedure for determining carbon black. Extract the residue remaining on the filtering crucible with hot acetone at 50° to 55°C using the filtrator connector to a water aspirator. Run approximately twenty (20) mL of hot acetone through the crucible without stirring until the crucible is warm. Wash the residue on the filter with five (5) washings of hot acetone using approximately twenty (20) mL per washing. The contact time of the washings to the residue in the crucible shall be five (5) to ten (10) seconds. Stir gently, during the washings, so as not to disturb the filter paper. Dry the crucible and its residue in an oven, at 95° + 5°C, for thirty (30) minutes. Cool in a desiccator for thirty (30) minutes, weigh and calculate the percent carbon black according to the following formula:

$$\text{Percent carbon black} = \frac{(D-E)\ 100}{C}$$

where:

D = weight of crucible plus second residue, in grams
C = weight of sample, in grams
E = weight of crucible, in grams

4.5.20.3 RDX determination. Calculate the percent RDX including nitrocellulose, as follows:

Percent RDX = 100 - (percent TNT + percent carbon black)

4.5.21 Workmanship. Take a portion of approximately 50g of the sample and spread it over a flat surface. Visually examine the material to determine compliance with 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. Carbon black shall be packed in fiber or metal drums. Fiber drums shall comply with the basic requirements for Type I, Grade A, Class 1 of PPP-D-723 except that they shall be of the locking band type with

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weather-resistant (hot melt adhesive) construction. Metal drums shall conform to Types IV, V or VI of PPP-D-705. Each drum shall be lined with a plastic bag conforming to PPP-D-26, for a Type I, Style 1, 2 or 3 bag. The interior bag shall be securely closed prior to closure of the drum. The gross weight of the container shall not exceed 200 pounds.

5.1.2 Level B. Twenty-five or fifty pounds, as specified in the contract or order, shall be packed in a shipping sack conforming to Numbers 1 or 1X of specification UU-S-48. Alternatively polyethylene bags complying with Package Number 623 of the Uniform Freight Classification may be used. Bag closures shall comply with the referenced bag specifications.

5.1.3 Commercial. Carbon black shall be packed in containers conforming to Uniform Freight Classification requirements and in accordance with the manufacturer's standard practice.

5.2 Marking. All containers shall be marked in accordance with MIL-STD-129. Marking shall include the following information:

a. Material Stock Number (if available)
b. Nomenclature of the item and grade.
c. Title, number and date of the Specification
d. Lot number and batch number
e. Date of manufacture
f. Contract number
g. Item trade name
h. Manufacturers name and address
i. Storage conditions

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6. NOTES

6.1 Intended use. Carbon black, covered by this specification, is intended for use in the manufacture of rocket propellant, stabilization of Cyclotol explosives, small arms propellants and liners.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.
b. Grade of carbon black required (see 1.2).
c. Whether the diphenylguanidine absorption requirement shall replace the oil absorption requirement for Grade A carbon black procurement (see 3.2).
d. Army requirements for fixed carbon and moisture content of Grade A carbon black, when applicable (see Table I).
e. First article sample requirements if other than specified in section 4 of this specification.
f. Quantity required and delivery schedules.
g. Level of packing, whether specific kind of container is required and net weight.

6.2.2 Contract data requirements

a. Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-E-1118 or DI-R-1714 with appropriate addendum to each). See 6.3.

b. Data cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the Equipment Lists referenced on the Equipment Tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. However, the contractor need not furnish any design when a complete Government SIE design is part of the Technical Data Package (TDP). Unless otherwise specified, the contractor

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may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract.

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The specific segment of ARRADCOM to which the Contractor designs shall be sent will be specified in the item detail specification. When the contractor submits inspection equipment designs to the Government for approval he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal.

Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Supersession. This specification includes the requirements of carbon black from the following specifications:

a) Grade D - Local specification 174-104, 16 April 1973, Carbolac I and Navy specification OS10008, Type III, 31 August 1959.
b) Grade E - Navy specification WS14382, 18 April 1973, Carbon Black.
c) Grade F - Navy specifications OS9434, 3 May 1963 and WS4967, 18 November 1964.

6.5 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.6 A Palo impeller agitator may be obtained from the Emil Greiner Scientific Corporation located at 22 N. Moore St., New York, New York (Cat. No. 77-939-01).

6.7 In order to calculate the Yield Value using units of the International Metric System, the following conversions are given:

1 dyne = 1×10^{-5} newtons (N)
1 centipoise = 1×10^{-3} Pascal seconds (Pa·s)

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6.8 Benzene is considered carcinogenic. Therefore, adequate ventilation must be provided when working with this solvent. A fume hood should be employed.

Custodians:
Army - AR
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Review Activities:
Army
Navy
Air Force
DSA Center - GS

Preparing Activity:
Army - AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-306 Carbon Black, Powdered, Dry

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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